MLLM-SHAP Research Direction Plan

Roadmap, Compute Demand, and Execution Timeline

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Talk map (10 min)

- Current plan status: blockers and execution gates
- Research tracks: audio/multimodal, text-to-text, extended program, infrastructure
- Compute demand model with explicit assumptions and dataset scales
- Integrated timeline with interleaved workstreams
- Scope for the master's thesis and conference papers
- Team extension

Research scope

- **Reproducibility requirement:** versioned artifacts, pinned environment, benchmark protocol.
- **T2T track:** estimand correctness, exact-vs-approx estimator validation, utility-function ablations.
- **SGPA track:** segmentation quality, masking ablations, faithfulness, stability, human validation.
- **Cross-model requirement:** primary model + secondary local model + secondary API model.
- **Cross-lingual requirement:** English + at least two typologically distinct languages.

Current blockers and execution risks

1. **Faithfulness evidence gap:** core faithfulness experiments are still incomplete and must be finished before strong claims.
2. **Hyperparameter-grid validation:** segmentation/masking hyperparameters need full validation on natural speech and multi-annotator protocol.
3. **Compute bottleneck:** planned experiment matrix is larger than single-GPU practical capacity; cluster scheduling is required.
4. **Rerun risk:** if ablations change the best masking strategy, previously reported results must be recomputed.

Research steps: text-to-text core

- **Phase 0:** immediate data integrity audit for estimand identity (Shapley vs Banzhaf) across existing T2T outputs.
- **Phase 1–2:** freeze 3-model text matrix and held-out prompt suite; enforce connector and masking invariants.
- **Phase 3:** synthetic toy games with exact ground truth plus convergence checks against a precise explainer reference.
- **Phase 4:** utility-function/scoring ablations (greedy similarity, log-probability, distribution-aware alternatives).
- **Phase 5–6:** faithfulness and robustness tests (AOPC/AUC, rank stability, FDR correction), then benchmark/reporting hardening.

Research steps: infrastructure and tooling

- **Data/artifacts:** move heavy results out of git, add artifact metadata (pipeline version, commit, config hash).
- **Framework outputs:** per-sample structured records with masks, responses, and hardware traces.
- **Performance:** optimize bottlenecks in masking/sampling and reduce RAM footprint for 10k+ coalition calls.
- **Reporting/visualization:** reproducible benchmark sidecars and notebook/GUI views for faithfulness and variance comparisons.
- **Role in program:** infrastructure track that enables reproducible scale-up of all scientific phases above.

Research steps: audio/multimodal core

- **Phase 0 (gate):** lock model matrix, resolve Shapley/Banzhaf labeling, formalize multimodal coalition game.
- **Phase 1:** verify additivity assumptions, retune segmentation hyperparameters on real speech with multi-annotator protocol, add audio preprocessing.
- **Phase 2:** CTC alignment benchmark vs naive baselines; masking strategy ablation (deletion vs silence vs ambient noise) as go/no-go gate.
- **Phase 3:** full-dataset execution, cross-lingual runs, replication on secondary models, and power-analysis-driven sample sizing.
- **Phase 4–6:** faithfulness/stability/K-sensitivity, human validation, theory/reproducibility, and explicit failure-mode characterization.

Research steps: extended research program

- **Phase 1:** multimodal interaction values (pairwise tractable SII-style studies), broader audio K-sensitivity, and plausibility/OOD metrics.
- **Phase 2:** generalization beyond current setup: encoder backbones, task families, gradient/attention baselines, model-size effects.
- **Phase 2:** modality trade-off analysis and fine-tuning impact on attribution rank structure.
- **Phase 3:** longer-horizon method work: learned coalition sampling, hierarchical grouping, KL-based scoring, contrastive explanations.
- **Phase 3:** input-space vs latent-space explanation layer comparison where internal representations are available.

Conference and thesis status

- **INTERSPEECH 2025**
  - SGPA method paper focused on spectrogram-guided phonetic alignment for feasible audio SV analysis.
  - Core content: feasibility bottleneck diagnosis, 4-stage SGPA pipeline, segmentation diagnostics, and evaluation reductions (calls/time) for audio explainability.
  - **Status:** rebuttal stage; likely rejection.
- **ACL 2026 System Demonstration**
  - `mllm-shap` platform/demo paper focused on system architecture and tooling.
  - Core content: multimodal masking, multi-turn token-role tracking, estimator suite (Exact/MC/CC/Neyman/Hierarchical), GUI workflow, and package-level benchmarking.
  - **Status:** accepted.
- **Bachelor thesis**
  - Thesis track aligned with the explainability platform and research program.
  - **Status:** final grade 5.

What is already evidenced in submitted work

- **From the SGPA conference paper:** demonstrated feasibility improvement for audio explainability, including strong reduction in model calls per sample and improved boundary-quality diagnostics.
- **From the ACL system demo paper:** end-to-end platform scope is already demonstrated (multimodal masking, estimator suite, GUI workflow, package integration).
- **Reviewer perspective (current signal):** engineering feasibility and system integration are seen as strengths; faithfulness depth, broader validation, and stronger replication are the main pressure points.
- **Planning implication:** next steps are scale-up, replication, and stricter validation, rather than proof-of-concept work.

Dataset sizes driving compute volume

Dataset block	Planned scope	Size
VoiceBench single-sentence subset	initial controlled English set with original + male TTS + female TTS variants	500
LibriSpeech single-sentence subset	initial controlled natural-speech English set	500
Full VoiceBench corpus	mandatory full run in large-scale phase	854
Full single-sentence corpus	mandatory full run in large-scale phase	448
Cross-lingual evaluation	at least two additional languages, minimum planned sample count	$\geq 2 \times 100$
Power-analysis pilot set	pilot for effect-size estimation before full commitments	20
Stability protocol set	repeated-seed and temperature protocols on fixed samples	20
K-sensitivity validation set	$K \in \{50, 100, 200, 500, 1000\}$ on fixed audio samples	20
Human-study set	user validation study for highlighted segments	$n \geq 15$ participants
T2T held-out prompt suite	6 prompt buckets (QA, extraction, rewrite, constrained gen, multi-turn, long-context)	50–100 per bucket

Lower-bound run count and runtime translation

Component	Derivation from plan minima	Sample-runs
Core full-corpus faithfulness matrix	1302×3 methods $\times 2$ full models	7812
Cross-lingual minimum matrix	$(2 \times 100) \times 3$ methods $\times 2$ full models	1200
Stability + K checks	$(20 \times 10) + (20 \times 5)$ from repeated seeds and K-sweep	300
Additional explicit subsets	30-sample independence + 100-sample granularity + 40 failure-mode cases	170
Audio lower bound	sum of explicit minima above (before secondary B and reruns)	**9482**

- **Definition:** one *sample-run* means one explained sample for one method-model configuration.
- **No-SGPA runtime mapping:** 9482 sample-runs $\approx$ 9482 GPU-hours $\approx$ 395 GPU-days (single GPU).
- **Mixed-runtime mapping:** if only the SGPA branch gets large speedup while native and LOO stay near baseline cost, total remains on the order of $\sim$ 6.4k GPU-hours.
- **T2T suite scale note:** 6 buckets $\times$(50 to 100) prompts gives 300 to 600 prompts; replicated across 2 research models and multiple explainers/scoring backends, this adds a substantial second compute block.
- **Lower-bound interpretation:** $\sim$ 10k GPU-hours covers only the SGPA-focused lower bound; remaining components and the T2T grid add further demand.

Timeline (interleaved by workstream)

Window	Primary focus and sequencing
Months 1–2	**Fix package and unblock execution:** connector stability, experiment metadata/reporting, estimand labeling, baseline reproducibility checks.
Months 3–5	**T2T full grid:** held-out prompt-suite matrix, explainer/scoring ablations, two-model replication, significance/FDR-ready outputs.
Months 5–8	**SGPA core:** segmentation and masking ablations, alignment benchmarking, full-corpus faithfulness/stability/K-sensitivity runs.
Months 8–10	**Multilingual and cross-model stage:** additional-language runs, TTS vs natural stratification, secondary-model verification of core rankings.
Months 10–13	**Multimodal extensions:** interaction-value studies, modality trade-off analysis, encoder/task generalization, advanced metrics beyond faithfulness.
Months 13–15	**Other/finalization:** failure-mode closure, theory appendix, reproducibility packaging, artifact governance, and performance optimization.

Risk register and mitigation

- **Connector/model instability** $\rightarrow$ lock fallback model candidates early.
- **Underpowered experiments** $\rightarrow$ pilot-based power analysis before full runs.
- **Circular faithfulness evaluation** $\rightarrow$ separate utility for SHAP computation vs evaluation.
- **Compute bottlenecks** $\rightarrow$ cluster queue planning and phase-gated high-information ablations.
- **Cross-model claim fragility** $\rightarrow$ explicit scoping if replication diverges.

Master's thesis and team needs

- Separate deliverables into thesis-oriented contributions and conference-paper contributions.
- Add dedicated DevOps support for cluster scheduling, artifact storage, and reproducible experiment orchestration.

Conference timeline and scope fit (May 2026 – Q1 2027)

Venue	Apply by	Scope link and expected deliverable	Fit
EMNLP 2026 main	May 25, 2026	Only viable if Paper 1 results are already in hand; otherwise forces premature writing	Low (skip)
BlackboxNLP 2026	$\sim$ mid-Aug 2026	Primary Paper 1 target: cross-modal grounding diagnostics with archival NLP interpretability audience	High
AAAI 2027	$\sim$ late Jul/early Aug 2026	Secondary/main-track backup for Paper 1 variant if results mature early	Medium
ICASSP 2027	$\sim$ early Sep 2026	SGPA plan-B venue and optional audio-flavored Paper 1 variant (4-page format)	High
NeurIPS workshops 2026	late Aug–mid Sep 2026	Short-form Paper 1 for additional audience exposure and feedback cycle	High
ICLR 2027	abs $\sim$ Sep 19; full $\sim$ Sep 24, 2026	Primary Paper 2 benchmark submission (SV/IG/attention/LIME/ablation suite)	Medium
ARR route (NAACL/EACL/ACL)	Oct 2026 / Feb 2027 cycles	Resubmission/escalation path after ICLR/workshop outcomes	Medium–High

Funding and compute applications

Target	Apply by	Role and gating dependency
PRELUDIUM 25	Jun 16, 2026	Feasible only with fast WUT host confirmation; otherwise skip and defer to next call
OPUS 31	Jun 16, 2026	High-leverage path if Maria leads PI role and proposal can be aligned quickly
FNP START 2027	Q4 2026 call	Prestige track aligned with accepted/archival publication outputs
LIDER UP 1/2027	Q1 2027 (eligibility-gated)	Requires doctoral-student or WUT employment status before call opens
Anthropic / OpenAI / Google credits	rolling (apply immediately)	Short-term API compute to de-risk benchmark and ablation execution
PLGrid / EuroHPC access	rolling/quarterly	Primary cluster backbone for high-volume phase 2–4 experiment matrices

- **Prioritization rule:** if bandwidth is constrained, deliver Paper 1 + SGPA plan B first; defer Paper 2.
- **Immediate actions (next 2 weeks):** apply for rolling compute credits and PLGrid access; finalize OPUS/PRELUDIUM decision with WUT stakeholders.